Vencislav Popov

August, 2026

Ven Popov | vencislav.popov@gmail.com | https://venpopov.com

Education

2020 Ph.D., Psychology, Carnegie Mellon University
Dissertation Title: Resource Depletion and Recovery in Human Memory
Adviser: Lynne M. Reder | Committee: John Anderson, Marlene Behrmann, Marc Coutanche
Glushko Dissertation Prize (2021)

2015 B.A., Psychology, New Bulgarian University

Research Experience

02.2024 - current Senior Scientist in Computational Modeling (tenured)
Department of Psychology, University of Zurich

08.2020 - 01.2024 Postdoctoral Researcher, Cognition Lab, University of Zurich

07.2015 - 06.2020 Doctoral Research, Memory Lab, Carnegie Mellon University

01.2015 - 08.2015 Data Analyst, Bulgarian Addictions Institute

Scholarships and Awards

2025 **Bertelson Early Career Award**
"To an outstanding young scientist for making a significant contribution to European Cognitive Psychology"
European Society for Cognitive Psychology

2023 **ESCOP Early Career Publication Award**
"For outstanding work that is truly transformational for the field of human learning and memory"
European Society for Cognitive Psychology

2021 **Robert J. Glushko Dissertation Prize ($10,000)**
For an outstanding dissertation in cognitive science
Cognitive Science Society & Robert J. Glushko and Pamela Samuelson Foundation

2020 **McClelland Outstanding Paper Award**
Center for the Neural Basis of Cognition, PA

Bobby Klatzky Graduate Student Publication Award
For outstanding performance in scholarly research and writing
Department of Psychology, Carnegie Mellon University

2019 **Bobby Klatzky Graduate Student Publication Award**
For outstanding performance in scholarly research and writing
Department of Psychology, Carnegie Mellon University

2017 **Cognitive Science Society Travel Award**
For an outstanding student paper
Robert J. Glushko and Pamela Samuelson Foundation

2015 - 2016 **Presidential Fellowship in the Life Sciences**
Richard King Mellon Foundation

Publications

(* = co-first author; † = corresponding author)

Preprints

Popov, V. (2026). A Poisson Count Race as a Generative Bridge Between Logit and Probit Choice. Preprint available at https://doi.org/10.31234/osf.io/r36vs_v1

Popov, V. (2026). *m*-Bell and *m*-Stirling numbers: Iterated binomial transforms, hyper-Bessel functions, and moments of the Conway–Maxwell–Poisson distribution. Preprint available at https://arxiv.org/abs/2608.12011

Popov, V. (2026). Cognitive resources can be intentionally released when processed information becomes irrelevant: Insights from the primacy effect in working memory. Preprint available at https://osf.io/preprints/psyarxiv/gct58

Barthelmes, B. T. & **Popov, V.** (2026). Towards measuring the theory crisis in psychology: An LLM-assisted discourse-analysis approach. Preprint available at https://doi.org/10.31234/osf.io/4y2sp_v2

van Dongen, N. N., Glöckner, A., Musfeld, P., Sarafoglou, A., Borsboom, D., **Popov, V.**, Yuan, L. (2026). The state and status of theory in psychological science. Preprint available at https://doi.org/10.31234/osf.io/2fjx4_v2

Greeno, D. J., Morey, C. C., & **Popov, V.** (2025). Encoding serial items depletes cognitive resource: An adversarial comparison of reconfiguration and resource-depletion accounts of working memory. Preprint available at https://doi.org/10.31234/osf.io/vj8pn_v3

Popov, V. & Oberauer, K. (2024). How to build an observatory for the mind. Preprint available at https://doi.org/10.31234/osf.io/hmu9r

Popov, V. (2023). If God Handed Us the Ground-Truth Theory of Memory, How Would We Recognize It? Preprint available at https://osf.io/preprints/psyarxiv/ay5cm

Journal publications

2026 Ma, S., **Popov, V.**, & Zhang, Q. (2026). Cognitive constraints and reward environments jointly shape memory formation. *Psychological Review.* Advance online publication. https://doi.org/10.1037/rev0000612

Dames, H., **Popov, V.**, & Oberauer, K. (2026). Forget for now, but remember later: Can people selectively remove information from working memory while keeping it in long-term memory? *Journal of Cognition, 9*(1), 25.

2025 Frischkorn, G.* & **Popov, V.* (co-first authors)** (2025). A tutorial for estimating Bayesian hierarchical mixture models for visual working memory tasks: Introducing the Bayesian Measurement Modeling (bmm) package for R. *Behavior Research Methods, 57*(5), 144.

Trueblood, J. S., Allison, D., Field, S. M., Fishbach, A., Gaillard, S. D. M., Gigerenzer, G., **Popov, V.**, Teodorescu, A. (2025). The misalignment of incentives in academic publishing and implications for journal reform. *Proceedings of the National Academy of Sciences, 122*(5), e2401231121.

Binz, M., Alaniz, S., Roskies, A., Aczel, B., Bergstrom, C. T., Allen, C., **Popov, V.** Schulz, E. (2025). How should the advancement of large language models affect the practice of science? *Proceedings of the National Academy of Sciences, 122*(5), e2401227121.

2024 Ma, S., **Popov, V.**, & Zhang, Q. (2024). A neural index reflecting the amount of cognitive resources available during memory encoding: A model-based approach. *Journal of Experimental Psychology: Learning, Memory & Cognition, 51*(3), 350–370.

Dames, H., Musfeld, P., **Popov, V.**, Oberauer, K., & Frischkorn, G. (2024). Responsible research assessment should prioritize theory development and testing over ticking open science boxes. *Meta-Psychology, 8*.

2023 Dames, H. & **Popov, V.** (2023). When does intent matter for memory? Bridging perspectives with Craik. *Journal of Experimental Psychology: General, 152*(11), 3300–3309.

Popov, V.† & Dames, H. (2023). Intent Matters: Resolving the Intentional vs Incidental Learning Paradox in Episodic Long-term Memory. *Journal of Experimental Psychology: General, 152(1), 268300*. Preprint available at https://osf.io/preprints/psyarxiv/jf2en

- 2023 ESCOP Early Career Publication Award

2022 Norton, C. M., Ibinson, J. W., Pcola, S. J., **Popov, V.**, Tremel, J. J., Reder, L. M., ... & Vogt, K. M. (2022). Neutral auditory words immediately followed by painful electric shock may show reduced next-day recollection. *Experimental Brain Research, 240*(11), 2939–2951.

Popov, V.†, So, M., & Reder, L. (2022). Memory resources recover gradually over time: The effects of word-frequency, presentation rate and list-composition on binding errors and mnemonic precision in source memory. *Journal of Experimental Psychology: Learning, Memory & Cognition, 48*(9), 1263–1280.

2021 Vogt, K., Ibinson, J., Smith, C., Citro, A., Norton, C., Karim, H., **Popov, V.**, Mahajan, A., Aizenstein, H., Reder, L. & Fiez, J. (2021). Midazolam and ketamine produce distinct neural changes in memory, pain, and fear networks during pain. *Anesthesiology, 135*(1), 69–82.

2020 **Popov, V.†**, & Reder, L. (2020). Greater discrimination difficulty during perceptual learning leads to stronger and more distinct representations. *Psychonomic Bulletin & Review, 27*(4), 768–775.

Psederska, E., Yankov, G. P., Bozgunov, K., **Popov, V.**, Vasilev, G., & Vassileva, J. (2020). Validation of the Levenson Self-Report Psychopathy Scale in Bulgarian Substance Dependent Individuals. *Frontiers in Psychology, 11*, 1110.

Popov, V.† & Reder, L.†(2020). Frequency effects on memory: A resource-limited theory. *Psychological Review. 127*(1), 1-46.

- 2020 McClelland Outstanding Paper Award, CNBC
- 2020 Bobby Klatzky Graduate Student Publication Award, CMU

2019 **Popov, V.†**, Marevic, I., Rummel, J., & Reder, L. (2019). Forgetting is a feature, not a bug: Intentionally forgetting some things helps us remember others by freeing up working memory resources. *Psychological Science. 30*(9), 1303-1317.

Popov, V.†, Zhang, Q., Koch, G. E., Calloway, R. C., & Coutanche, M. N. (2019). Semantic knowledge influences whether novel episodic associations are represented symmetrically or asymmetrically. *Memory & Cognition. 47*(8), 1567-1581.

2018 **Popov, V.†**, Ostarek, M. & Tenison, C. (2018). Practices and pitfalls in inferring neural representations. *NeuroImage. 174*(1), 340-351

- 2019 Bobby Klatzky Graduate Student Publication Award, CMU

Shen, Z.*, **Popov, V.* (co-first authors)**, Delahay, A., & Reder, L. (2018). Item strength affects working memory capacity. *Memory & Cognition. 46*(2), 204-215.

2017 **Popov, V.†**, Hristova, P., & Anders, R. (2017). The Relational Luring Effect: Retrieval of relational information during associative recognition. *Journal of Experimental Psychology: General. 146*(5), 722-745

Manelis, A.*†, **Popov, V.*† (co-first authors)**, Paynter, C., Walsh, M., Wheeler, M., Vogt, K., & Reder, L.† (2017). Cortical networks involved in memory for temporal order. *Journal of Cognitive Neuroscience. 29*(7), 1253-1266.

2016 Reder, L. M., Liu, X. L., Keinath, A., **Popov, V.** (2016). Building knowledge requires bricks, not sand: The critical role of familiar constituents in learning. *Psychonomic Bulletin & Review*. *23*(1), 271-277.

2015 **Popov, V.†** & Hristova, P. (2015). Unintentional and efficient relational priming. *Memory & Cognition*, *43*(6), 866-878.

Book chapters

2024 **Popov, V.** & Reder, L. (2024). Frequency effects in recognition and recall. In M. Kahana & A. Wagner (Eds.), *The Oxford Handbook of Human Memory* (pp. 765–796). Oxford University Press. https://doi.org/10.1093/oxfordhb/9780190917982.013.27

Refereed full conference papers

2018 Zhang, Q.*, **Popov, V.* (co-first authors)**, Koch, G. E., Calloway, R. C., & Coutanche, M. N. (2018). Fast Memory Integration Facilitated by Schema Consistency. In C. Kalish, M. Rau, J. Zhu, T. Rogers (Eds.), *Proceedings of the 40th Annual Conference of the Cognitive Science Society*. Austin, TX: Cognitive Science Society.

2017 **Popov V.**, Ostarek, M., & Tenison, C. (2017). Inferential Pitfalls in Decoding Neural representations. In G. Gunzelmann, A. Howes, T. Tenbrink, & E. Davelaar (Eds.), *Proceedings of the 39th Annual Conference of the Cognitive Science Society* (pp. 961-966). Austin, TX: CSS.

Popov V., & Reder, L. (2017). Repetition improves memory by strengthening existing traces: Evidence from paired-associate learning under midazolam. In G. Gunzelmann, A. Howes, T. Tenbrink, & E. Davelaar (Eds.), *Proceedings of the 39th Annual Conference of the Cognitive Science Society* (pp. 2913-2918). Austin, TX: Cognitive Science Society.

Popov V., & Reder, L. (2017). Target-to-distractor similarity can help visual search performance. In G. Gunzelmann, A. Howes, T. Tenbrink, & E. Davelaar (Eds.), *Proceedings of the 39th Annual Conference of the Cognitive Science Society* (pp. 968-973). Austin, TX: Cognitive Science Society.

2014 **Popov, V.** & Hristova, P. (2014). Automatic analogical reasoning underlies structural priming in comprehension of ambiguous sentences. In P. Bello, M. Guarini, M. McShane, & B. Scassellati (Eds.), *Proceedings of the 36th Annual Conference of the Cognitive Science Society* (pp. 1192-1197). Austin, TX: Cognitive Science Society.

Popov, V. & Petkov, G. (2014). The level of processing affects the magnitude of induced retrograde amnesia. In P. Bello, M. Guarini, M. McShane, & B. Scassellati (Eds.), *Proceedings of the 36th Annual Conference of the Cognitive Science Society* (pp. 2787-2792). Austin, TX: CSS.

Journal publications (in Bulgarian)

2016 **Popov, V.**, Nedelchev, D., Peneva, E., Psederska, E., Georgieva, V., Vasilev, G., & Vassileva, J. (2016). Psychometric Characteristics of the Bulgarian Version of the Buss-Warren Aggression Questionnaire (BWAQ). *Clinical and Consulting Psychology, 4*(30), 37-53.

Nedelchev, D., **Popov, V.**, Psederska, E., Bozgunov, K., Vasilev, G., Peneva, E., & Vassileva, J. (2016). Psychometric Characteristics of the Bulgarian Version of the Wender Utah Rating Scale (WURS-25) for ADHD. *Clinical and Consulting Psychology, 8*(2), 3-17 (in Bulgarian)

Popov, V., Psederska, E., Peneva, E., Bozgunov, K., Vasilev, G., Nedelchev, D., & Vassileva, J. (2016). Psychometric Characteristics of the Bulgarian Version of the Toronto Alexithymia Scale (TAS-20). *Psychological Research, 19*(2), 25-42. (in Bulgarian)

2015 **Popov, V.**, Bozgunov, K., Vasilev, G., & Vassileva, J. (2015). Psychometric Characteristics of the Bulgarian Version of Levenson's Self-report Psychopathy Scale. *Bulgarian Journal of Psychology, 1-4*, 253-278. (in Bulgarian)

Open-source software

bmm Lead author. An R package for Bayesian Measurement Modeling of Visual Working Memory

Presentations

Invited talks

2026 Keynote, SciLib Forum 2026: AI or Human in the Loop?, University Medical Library Basel, Switzerland, Sep 2026
AI as a Thinking Partner: The Extended Mind in Scientific Practice

2025 Bertelson Award Keynote Lecture, ESCoP, Sheffield, UK, Sep 2025
Psychology as the Computational Modeller Views It

2023 One World Cognitive Psychology Seminar Series, Psychonomic Society, online, Sep 2023
If God Handed Us the Ground-Truth Theory of Memory, How Would We Recognize It?

2022 Cognitive Brown Bag Series, Michigan State University, online, November 2022
Resource Depletion and Recovery in Human Memory

Working Memory Development Lab Seminar, University of Fribourg, April 2022
Frequency effects on memory: A resource-limited theory

2021 Context and Episodic Memory Symposium, University of Melbourne, online, Dec 2021
Intent Matters: Resolving the Intentional vs Incidental Learning Paradox in Episodic Long-term Memory

Gluschko Dissertation Award Symposium, Cognitive Science Society, online, July 2021
Frequency effects on memory: A resource-limited theory

Seminar on Memory and Analogy, New Bulgarian University, online, May 2021
Intent Matters: Resolving the Intentional vs Incidental Learning Paradox in Episodic Long-term Memory

2019 Cognitive Seminar, Ohio State University, Columbus, OH, April 2019
Semantic relations play a crucial role in memory, language comprehension and analogical reasoning

2016 Workshop on Memory and Skill, Duke University, Durham, North Carolina, April 2016.
Not knowing what we know: A call for a theory-neutral database for empirical results in psychology.

Departmental talks

2019 Visual Cognition Meeting (VisCog), Carnegie Mellon University, Pittsburgh, Jan 2019. *Practices and pitfalls in inferring neural representations*.

2017 CNBC Brain Bag series, Center for the Neural Basis of Cognition, Pittsburgh, Nov 2017. *Inferential Pitfalls in Decoding Neural Representations*.

Perception, Action and Learning (PAL) series, Carnegie Mellon University, Pittsburgh, Feb 2017. *The Mortar of Cognition: Semantic relations play a crucial role in memory, language comprehension and analogical reasoning.*

Conference talks

2026 Can artificial agents build real culture? *Oral presentation at the Annual Summer Interdisciplinary Conference (ASIC 2026)*. Squamish, BC, Canada

2023 If God Handed Us the Ground-Truth Theory of Memory, How Would We Recognize It? *Oral presentation at the 23rd Annual Summer Interdisciplinary Conference (ASIC 2023)*. Kranjska Gora, Slovenia

2021 Uncertainty ratings can improve the estimation of memory precision by several orders of magnitude. *Oral presentation at the Psychonomic Society 62nd Annual Meeting*. Virtual conference.

2019 Word frequency affects binding probability not memory precision. *Oral presentation at the 41st Annual Conference of the Cognitive Science Society*. Montreal, CA.

The effect of semantic relatedness on associative symmetry in memory. *Oral presentation at the 41st Annual Conference of the Cognitive Science Society*. Montreal, CA.

Frequency effects on memory: A resource-limited theory. *Oral presentation at the 18th Annual Summer Interdisciplinary Conference (ASIC 2019)*. Seefeld, Austria

2017 Inferential Pitfalls in Decoding Neural Representations. *Oral presentation at the 39th Annual Conference of the Cognitive Science Society*. London, UK.

The Relational Luring Effect: False Recognition via Relational Similarity. *Oral presentation at the 39th Annual Conference of the Cognitive Science Society*. London, UK.

Target-to-distractor similarity can help visual search performance. *Oral presentation at the 39th Annual Conference of the Cognitive Science Society*. London, UK.

The Relational Luring Effect: Retrieval of relational information during associative recognition. *Oral presentation at the 16th Annual Summer Interdisciplinary Conference (ASIC 2017)*. Interlaken, Switzerland.

2015 Unintentional relational priming. *Oral presentation at the Annual Conference of the Department of Cognitive Science and Psychology*. Kazanlak, Bulgaria

2014 Priming of relations: Unintended, unconscious and efficient. *Oral presentation during the European Campus of Excellence Summer School in Memory and Mind: Perspectives from Philosophy and Neuroscience.* Germany, Bochum: Ruhr University Bochum

Automatic analogical reasoning underlies structural priming in comprehension of ambiguous sentences. *Oral presentation at the 36th Annual Conference of the Cognitive Science Society*. Quebec City, Canada.

Retrograde amnesia as a function of the level of processing. *Oral presentation at the Annual Conference of the Department of Cognitive Science and Psychology*. Belmeken, Bulgaria: NBU

Automatic analogies in language comprehension. *Oral presentation at the Annual Conference of the Department of Cognitive Science and Psychology*. Belmeken, Bulgaria: NBU

2013 Experiencing failure increases problem-solving time in current performance. *Oral presentation at the Annual Conference of the Department of Cognitive Science and Psychology*. Ribaritza, Bulgaria

Teaching experience

Mentorship

Berit Barthelmes, UZH, Master's thesis supervisor
Huichao Ji, UZH, Master's internship (now a PhD student at Yale University)
Lorin Schöni, UZH, Master's internship (now a PhD student at ETH Zurich)

Bachelor's thesis supervision at the University of Zürich

2026 Pablo Yac Barragán Reyes
Sabrina Silva
Yuzan Flury
2025 Aaron Steinhübl
Ida Brigitte Baumgardt
Noël Napoli
2022 Andjela Jovanovic
Lennart Müller

Graduate courses at the University of Zürich

Introduction to Computational Modeling of Behavior
Fall 2024, Fall 2026

Bayesian Statistics with Stan and brms
Spring 2025, Spring 2027

Introduction to Programming and Scientific Computing
Spring 2025, Spring 2026, Spring 2027

Generating and Testing Explanations of Brain and Behavior: The Good, the Bad, and the Ugly
(overall course satisfaction 5.0/6.0; satisfaction with the instructor 5.5/6.0)
Spring 2022, Spring 2023, Fall 2024, Fall 2026 (doctoral program)

The Psychology of Memory
(overall course satisfaction 4.3/6.0; satisfaction with the instructor 5.2/6.0)
Fall 2023

How do we measure cognition? (co-teaching with Hannah Dames)
(overall course satisfaction 5.6/6.0; satisfaction with the instructor 5.9/6.0)
Fall 2022

Data Science for Psychologists (online)
(overall course satisfaction 4.0/6.0; satisfaction with the instructor 5.0/6.0)
Spring 2021

Undergraduate courses at the University of Zürich

Cognitive Psychology 1 (lecture; co-taught with Lea Bartsch in 2024)
(2024: overall course satisfaction 4.2/6.0; satisfaction with the instructor 4.8/6.0)
Spring 2024, Spring 2026

Teaching assistant

2019 Cognitive Research Methods (Instructor: Erik Thiessen)
2018 Introduction to Psychology (Instructor: Danniel Oppenheimer)
Data Science for Psychology & Neuroscience (Instructor: Timothy Verstynen)
2017 Introduction to Parallel Distributed Processing (Instructor: David Plaut)

Guest lectures

2019 Computational modelling of memory (Memory. Instructor: Lynne Reder)
2018 Approaches to studying memory (Memory. Instructor: Lynne Reder)
2018 Introduction to classifiers (Data Science for Psychology & Neuroscience. Instructor: Timothy Verstynen)
2015 Working Memory (Memory. Instructor: Penka Hristova)
2014 Working Memory (Memory and meaning. Instructor: Encho Gerganov)

Non-academic teaching

2022 Alpine Skill-transfer Workshops Coordinator (Academic Alpine Club of Zurich)
2019 Mountaineering School Director (Explorer's Club of Pittsburgh)
2019 Rock Climbing School Instructor (Explorer's Club of Pittsburgh)

Editorial and service roles

2021 - current Consulting Editor, *Memory and Cognition*
2021 - current Symposium co-organizer, *Distributed Working Memory Series*

Ad hoc reviewer

- Proceedings of the National Academy of Sciences
- NeuroImage
- Psychological Review
- Psychonomic Bulletin & Review
- Memory and Cognition
- Journal of Experimental Psychology: General
- Journal of Experimental Psychology: Learning, Memory & Cognition
- Cognitive Psychology
- Quarterly Journal of Experimental Psychology
- Memory
- Attention, Perception & Psychophysics
- Developmental Psychology
- Cognition
- PLOS One
- Behavior Research Methods
- Cognitive Science
- Cognitive Processing
- Brain & Cognition
- Experimental Psychology
- Communications Biology
- Scientific Reports
- Neural Computation
- Journal of Psycholinguistic Research
- Cognitive Science Society Conference